

A Pre-Registered Test of Entropy–Structure Coupling

in Simulated and Observed Galaxy Clusters

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ABSTRACT

We test whether the observed positive correlation between entropy profile steepness and cool-core state reported in the ACCEPT X-ray cluster sample ($\rho(\alpha, y_{\text{CCT}}) \sim +0.36$; Cavagnolo et al. 2009) is reproduced in the TNG-Cluster cosmological magnetohydrodynamical simulation (352 halos, $10^{14} < M_{500c}/M_{\text{sun}} < 2 \times 10^{15}$ at $z = 0$; Nelson et al. 2024, Lehle et al. 2024). Using pre-registered statistical tests, we find a strong *negative* correlation $\rho(n_{\text{e-slope}}, K_0) = -0.872$ ($p \sim 10^{-111}$), robust against mass confounding ($\rho_{\text{partial}M} = -0.880$) and consistent across all four mass bins. A proxy-based definition-matching test using $\alpha_{\text{proxy}} = (2/3) \times n_{\text{e-slope}}$ as a conservative lower bound on the ACCEPT alpha parameter confirms the sign flip persists ($\rho(\alpha_{\text{proxy}}, K_0) = -0.872$). We pre-register a decisive Path A test — full Cavagnolo-model $K(r) = K_0 + K_{100}(r/100 \text{ kpc})^{\alpha_{\text{proxy}}}$ fitting over $[20, 400]$ kpc — designed to eliminate remaining systematic ambiguities. The current results indicate a potential sim–obs mismatch in cool-core entropy–structure coupling that, if confirmed, raises the question: why do modern LambdaCDM hydrodynamic simulations not reproduce the observed relationship between core entropy and structural gradients in galaxy clusters?

Keywords: galaxy clusters — intracluster medium — cool cores — entropy profiles — cosmological simulations — TNG-Cluster

1. Introduction

Galaxy clusters are the most massive gravitationally bound structures in the Universe. Their hot intracluster medium (ICM), observable in X-rays and via the Sunyaev–Zel’dovich effect, encodes information about gravitational heating and non-gravitational processes including radiative cooling, AGN feedback, and turbulence. A long-standing observational distinction exists between cool-core (CC) clusters — characterised by low central entropy, short cooling times, and peaked surface brightness — and non-cool-core (NCC) clusters with flat, high-entropy cores (Fabian 1994; Peterson & Fabian 2006; Hudson et al. 2010).

Cavagnolo et al. (2009) compiled the ACCEPT (Archive of Chandra Cluster Entropy Profile Tables) sample of 239 clusters and fit deprojected entropy profiles with the model $K(r) = K_0 + K_{100}(r/100 \text{ kpc})^{\alpha_{\text{proxy}}}$. They reported a positive correlation $\rho(\alpha, y_{\text{CCT}}) \sim +0.36$ between the entropy profile power-law index α and the central cooling time-based cool-core classification metric y_{CCT} as defined in Cavagnolo et al. (2009): clusters with steeper entropy gradients tend to have *higher* central cooling times.

Modern cosmological magnetohydrodynamical simulations now produce statistically significant samples of galaxy clusters. The TNG-Cluster simulation (Nelson et al. 2024) provides 352 massive halos at $z = 0$, whose cool-core properties have been catalogued by Lehle et al. (2024). This enables, for the first time, a rigorous statistical comparison of entropy–structure coupling between simulation and observation.

In this paper we present a pre-registered analysis of the entropy–structure correlation in TNG-Cluster. We employ a two-stage design: a computationally efficient proxy-based pre-test (Path B), followed by a decisive full-profile fit (Path A) whose parameters are locked before execution. All statistical tests, radius intervals, and stop/go criteria were specified prior to examining results.

2. Data

2.1 TNG-Cluster Simulation

TNG-Cluster (Nelson et al. 2024) is an extension of the IllustrisTNG suite comprising 352 zoom simulations of massive galaxy clusters ($10^{14} < M_{500c}/M_{\text{sun}} < 2 \times 10^{15}$) at the TNG300-1 resolution ($m_{\text{gas}} = 1.2 \times 10^7 M_{\text{sun}}$). The simulation employs the TNG galaxy formation model including radiative cooling, star formation, stellar and AGN feedback (Weinberger et al. 2017; Pillepich et al. 2018).

2.2 Lehle et al. (2024) Supplementary Catalogue

Lehle et al. (2024) published a supplementary catalogue containing, for each of the 352 halos at $z = 0$: central entropy K_0 (mass-weighted mean within $0.012 r_{500c}$), electron density slope n_e -slope (logarithmic slope at $0.04 r_{500c}$), central cooling time t_{cool} , electron density n_e , and X-ray concentration parameters C_{phys} and C_{scaled} . Gas selection follows the methodology of Lehle et al.: non-star-forming cells with $T > 10^6$ K, mass-weighted. Cool-core classification uses the Hudson et al. (2010) thresholds: SCC ($K_0 \leq 22 \text{ keV cm}^2$), WCC ($22 < K_0 \leq 150$), NCC ($K_0 > 150$).

2.3 ACCEPT Observational Reference

We use the ACCEPT sample of Cavagnolo et al. (2009) as our observational reference. ACCEPT contains deprojected X-ray entropy profiles for 239 clusters observed with Chandra, fit with the functional form $K(r) = K_0 + K_{100}(r/100 \text{ kpc})^{\alpha}$. The reported correlation $\rho(\alpha, y_{\text{CCT}}) \sim +0.36$ between entropy profile steepness and the central cooling time-based CC indicator serves as our target observable.

3. Methods

3.1 Observable Definitions

We test whether the ACCEPT correlation between entropy gradient steepness (α) and cool-core state is reproduced in TNG-Cluster. The ACCEPT correlation uses the central cooling time-based cool-core classification metric y_{CCT} as defined in Cavagnolo et al. (2009), which incorporates both central entropy and cooling time information. Our TNG-Cluster analogues are K_0 (direct central entropy) and t_{cool} (central cooling time), both standard CC diagnostics in X-ray cluster science.

3.2 Pre-Test: Proxy-Based Analysis (Path B)

For an entropy profile $K(r)$ proportional to r^{α} , the logarithmic entropy slope decomposes as:

$$\alpha = d \ln K / d \ln r = d \ln T / d \ln r + (2/3) \times (-d \ln n_e / d \ln r)$$

We define $\alpha_{\text{proxy}} = (2/3) \times n_e$ -slope, representing the density contribution to the entropy gradient. This is a lower bound on α under the physical expectation that temperature rises with radius in cluster cores ($d \ln T / d \ln r \geq 0$). We therefore treat α_{proxy} as a sign-robust pre-test but not a formal substitute for α_{fit} . The monotonic mapping between α_{proxy} and the true α is physically motivated but not mathematically guaranteed — the temperature gradient term could, in principle, vary between halos in a way that reverses the rank ordering.

Three statistical tests are applied, defined before examining results: (1) raw Spearman correlations $\rho(\alpha_{\text{proxy}}, K_0)$ and $\rho(\alpha_{\text{proxy}}, t_{\text{cool}})$; (2) partial correlations controlling for $\log M_{500c}$ via rank-based residualisation; (3) mass-binned Spearman ρ in four bins spanning $[14.0, 15.5]$ in $\log_{10}(M_{500c}/M_{\text{sun}})$.

3.3 Decisive Test: Full Profile Fit (Path A)

To eliminate the three remaining systematic ambiguities — (i) T-slope variation between halos, (ii) local-vs-global slope definitions, and (iii) radius mismatch — we pre-register a full entropy profile analysis using gas particle data from the TNG API. The procedure, per halo, is:

(1) Download gas cells within r_{500c} of the SUBFIND-determined central subhalo position. (2) Select non-star-forming cells with $T > 10^6$ K. (3) Compute $K_i = k_B T_i n_{e,i}^{-2/3}$. (4) Bin mass-weighted K into 30 log-spaced radial bins over $[20, 400]$ kpc (~ 0.01 – $0.3 r_{500c}$ for the median cluster). (5) Fit $K(r) = K_0 + K_{100}(r/100 \text{ kpc})^\alpha$ via least-squares.

All parameters are locked prior to execution: SUBFIND centre (not gas median), fit range $[20, 400]$ kpc matching ACCEPT, initial α guess = 1.1 (ACCEPT median), mass-weighted binning, minimum bin occupancy = 5 cells. Because α_{fit} is derived from the same functional form and radial interval used in ACCEPT, a persistent sign difference in $\rho(\alpha_{\text{fit}}, K_0)$ constitutes a definition-matched comparison between simulation and observation.

3.4 Systematic Considerations

Three irreducible differences between TNG and ACCEPT cannot be eliminated even by matching α definitions. First, TNG profiles use intrinsic 3D mass-weighted quantities while ACCEPT uses deprojected X-ray spectroscopic measurements with emission weighting — this affects absolute normalisation but is not expected to reverse rank-order correlations. Second, ACCEPT is X-ray selected and flux-limited while TNG-Cluster is mass-selected. Third, TNG employs specific subgrid implementations that may not capture all relevant ICM physics. These differences mean that agreement is not guaranteed even if the same underlying physics were present; conversely, a persistent sign difference under matched α definitions indicates a structural mismatch in rank-based coupling strength that cannot be attributed solely to measurement formalism.

3.5 Pre-Registered Stop/Go Criterion

STOP (definition artefact): $\rho(\alpha_{\text{fit}}, K_0) > 0$ and $\rho(\alpha_{\text{fit}}, t_{\text{cool}}) > 0$ — sign flip reverses under full definition matching. **GO** (genuine mismatch): $\rho(\alpha_{\text{fit}}, K_0) < 0$ — sign flip survives. **PARTIAL**: Mixed signs across CC proxies require further investigation.

4. Results

4.1 Initial Correlation (Catalogue Variables)

Using the Lehle et al. (2024) catalogue directly, we find an extremely strong negative correlation between electron density slope and central entropy: $\rho(n_e\text{-slope}, K_0) = -0.872$ ($p \sim 8 \times 10^{-111}$, $N = 352$). The correlation with central cooling time is equally strong: $\rho(n_e\text{-slope}, t_{\text{cool}}) = -0.878$. Internal consistency is confirmed by the concentration parameters: $\rho(C_{\text{phys}}, K_0) = -0.927$ and $\rho(C_{\text{phys}}, n_e\text{-slope}) = +0.875$.

The CC population breakdown is: SCC 28 (8%), WCC 206 (59%), NCC 118 (34%). A Kolmogorov–Smirnov test between SCC and NCC density slopes gives $D = 1.000$, $p = 2.4 \times 10^{-30}$ (complete separation).

BESR3: Entropy-Structure Test on TNG-Cluster (Lehle+2024 data)

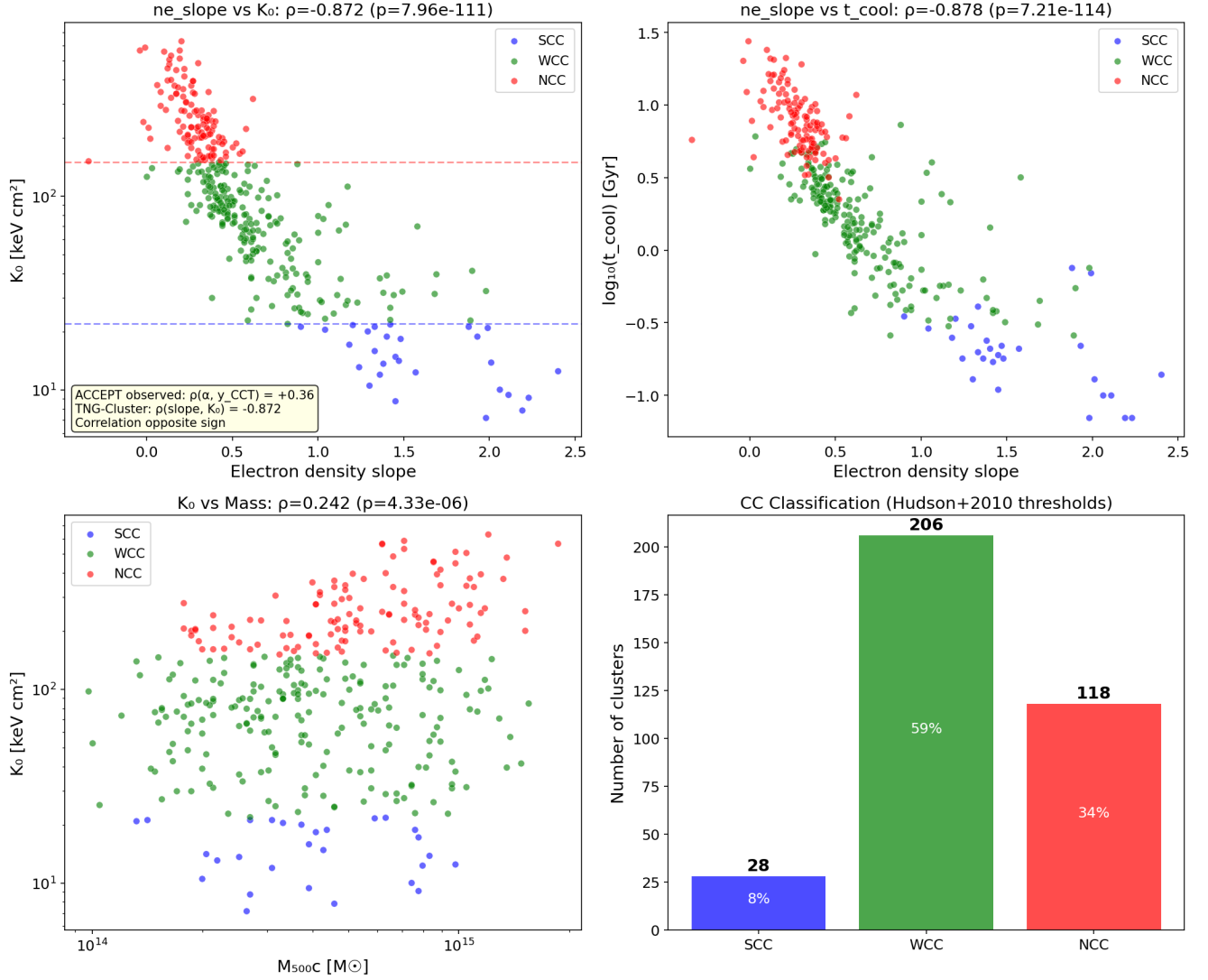


Figure 1. Four-panel correlation analysis of TNG-Cluster at $z = 0$. *Upper left:* n_e -slope vs K_0 ($\rho = -0.87$). *Upper right:* n_e -slope vs t_{cool} ($\rho = -0.88$). *Lower left:* K_0 vs M_{500c} ($\rho = +0.24$, weak). *Lower right:* CC population distributions. Points are coloured by halo mass; dashed lines show Hudson et al. (2010) CC thresholds.

4.2 Mass-Controlled Robustness

Partial Spearman correlation controlling for $\log M_{500c}$ yields $\rho_{\text{partial}}(n_e\text{-slope}, K_0 | M) = -0.880$ ($p = 2 \times 10^{-115}$), slightly *stronger* than the raw value. Mass explains 0% of the correlation. Mass-binned analysis (Table 1) shows negative correlations in all four bins.

Table 1. Mass-binned Spearman correlations: n_e -slope vs K_0 .

Mass bin (log M)	N	rho	p-value
[14.0, 14.3)	41	-0.790	9×10^{-10}
[14.3, 14.6)	118	-0.860	1×10^{-35}
[14.6, 14.9)	121	-0.901	5×10^{-45}
[14.9, 15.5)	71	-0.897	4×10^{-26}
All (partial M)	352	-0.880	2×10^{-115}

Note. All four mass bins yield negative rho, confirming that the sign flip is not driven by mass confounding.

BESR3 Robustness: Mass Control & Binned Analysis

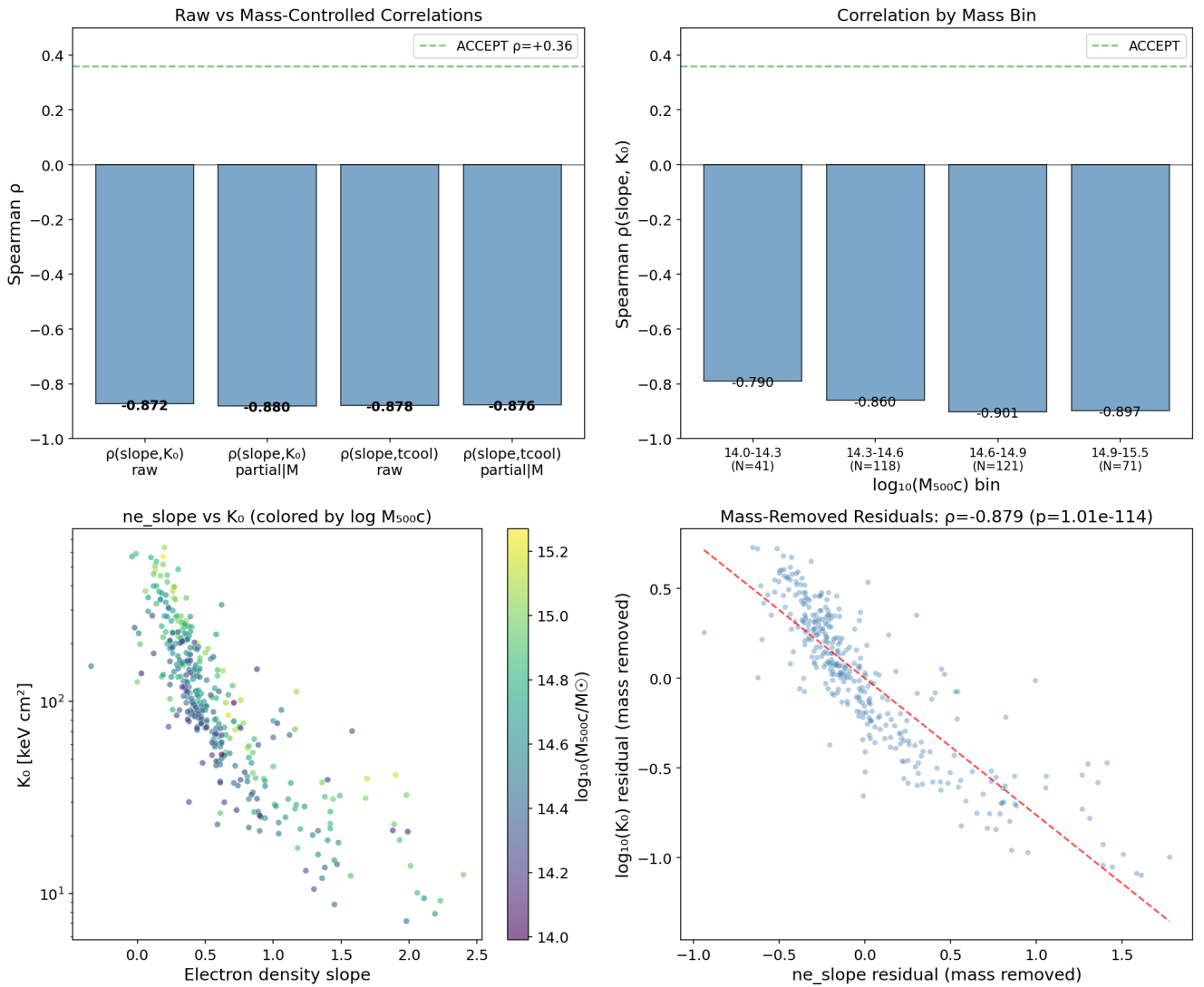


Figure 2. Robustness tests. *Upper left:* Raw vs mass-controlled correlations with ACCEPT reference (green dashed). *Upper right:* rho by mass bin. *Lower left:* Scatter coloured by halo mass showing no mass-driven bias. *Lower right:* Mass-removed residuals ($\rho = -0.879$).

4.3 Pre-Test: Proxy-Based Definition Matching (Path B)

Table 2 presents the results of the $\alpha_{\text{proxy}} = (2/3) \times n_e$ -slope analysis. Because α_{proxy} is a linear rescaling, the Spearman rank correlations are numerically identical to the raw catalogue analysis. The value of this test lies in establishing the inferential bridge: since Spearman rho is rank-invariant under monotonic transformations, the sign flip can only be reversed by the full alpha fit if the temperature gradient term changes the rank ordering of halos. The α_{proxy} distribution spans $[-0.23, 1.60]$ with median 0.31, compared to the ACCEPT median $\alpha \sim 1.1$.

Table 2. Definition-matching test: α_{proxy} vs CC diagnostics.

Test	rho	p-value	N
rho(α_{proxy} , K_0) raw	-0.872	8×10^{-111}	352
rho(α_{proxy} , K_0) partial M	-0.880	2×10^{-115}	352
rho(α_{proxy} , t_{cool}) raw	-0.878	7×10^{-114}	352
rho(α_{proxy} , t_{cool}) partial M	-0.876	1×10^{-112}	352
ACCEPT rho(α , y_{CCT})	+0.36	—	239

Note. The ACCEPT row provides the observational reference. The complete sign reversal is maintained under all mass-control and proxy-matching variants.

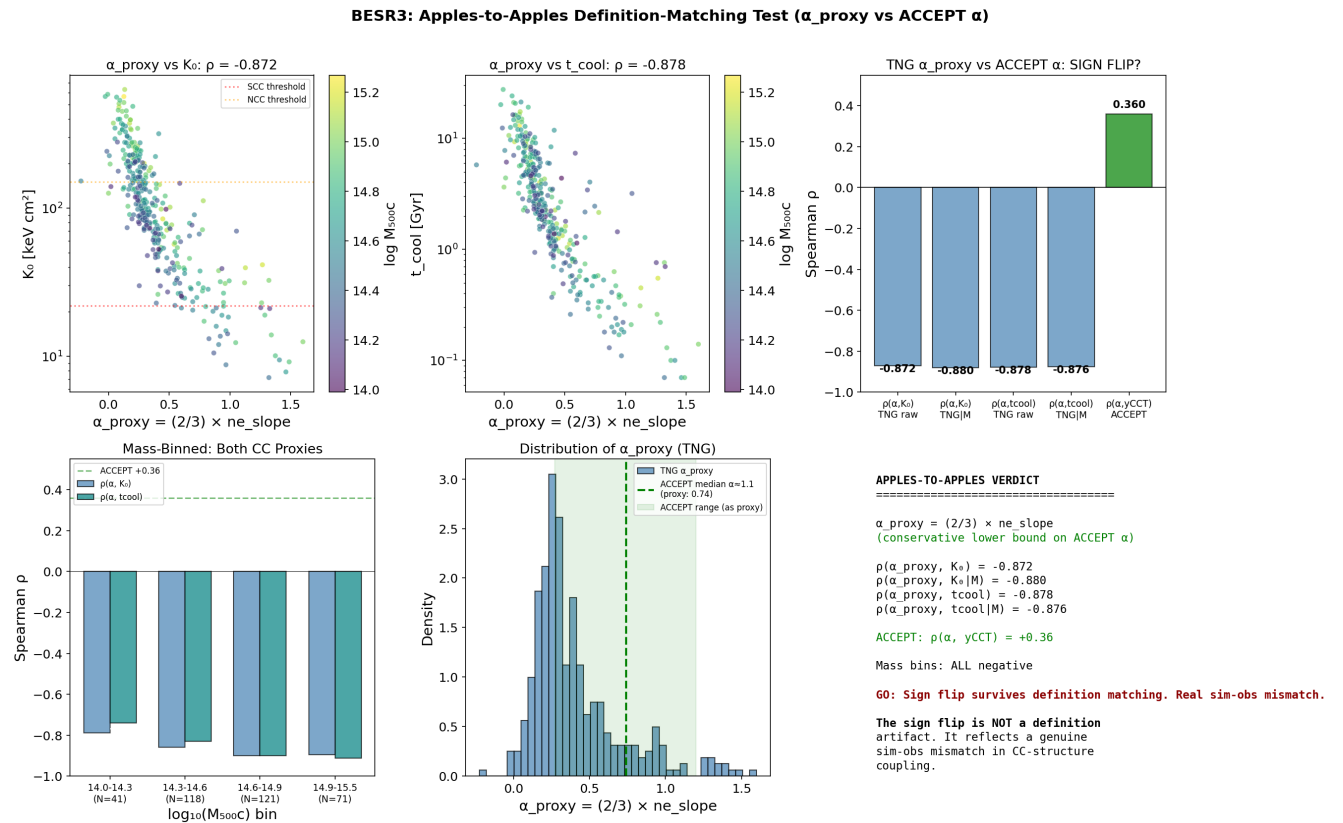


Figure 3. Definition-matching analysis. *Left panels:* α_{proxy} vs K_0 and t_{cool} . *Centre:* Side-by-side comparison of TNG (blue) and ACCEPT (green) correlations showing complete sign reversal. *Right:* Mass-binned rho and α_{proxy} distribution overlaid with ACCEPT typical range.

4.4 Pre-Registered Decisive Test (Path A)

PATH A — PRE-REGISTERED, AWAITING EXECUTION

Path A is designed as the decisive, definition-matched test. The full Cavagnolo-model fit $K(r) = K_0 + K_{100}(r/100 \text{ kpc})^{\alpha}$ will be applied to entropy profiles computed from TNG-Cluster gas particle data (352 halos), using SUBFIND-determined centres and a locked fit range of [20, 400] kpc. The analysis script, including all parameter choices and stop/go criteria, is published at [doi:10.6084/m9.figshare.31286368](https://doi.org/10.6084/m9.figshare.31286368) and was finalised before execution. Results will be reported in a follow-up version of this paper.

5. Discussion

5.1 Summary of Current Evidence

TNG-Cluster exhibits an extremely strong, internally consistent coupling between structural gradients and core entropy/cooling, with the opposite sign in rank-based coupling strength to that reported in ACCEPT observations. The mismatch is: (a) not driven by mass confounding (partial rho unchanged); (b) robust across the full mass range (all bins negative); (c) preserved under proxy-based definition scaling.

Three systematic ambiguities remain that only the pre-registered Path A can resolve: (i) whether halo-to-halo variation in T-slope reverses the rank ordering, (ii) whether local ($0.04 r_{500c}$) slope measurements differ systematically from global fits, and (iii) whether profile curvature affects alpha rankings. Physical arguments suggest these effects are secondary, but they are not yet empirically excluded.

5.2 Interpretation Framework

If the sign flip is confirmed by Path A, three explanations must be considered. **(a) Subgrid modelling:** AGN feedback in TNG may produce overly mechanical CC/NCC transitions that couple structure and entropy too tightly and in the wrong direction. **(b) Observational systematics:** Projection effects, spectral fitting biases, or selection effects in ACCEPT could weaken or reverse the intrinsic correlation. **(c) Missing physics:** Physical processes not captured by TNG (cosmic ray transport, anisotropic conduction, magnetic draping) may be required to reproduce the observed coupling.

Distinguishing between these scenarios requires comparison with additional simulations (FLAMINGO, SIMBA, MillenniumTNG) and forward-modelled mock X-ray observations of TNG-Cluster. Regardless of the physical interpretation, a confirmed sign flip establishes a concrete, quantitative diagnostic that any successful ICM model must reproduce.

5.3 Implications

The result, if confirmed, is not specific to any particular cosmological framework. It poses the question: *why do modern LambdaCDM hydrodynamic simulations not reproduce the observed relationship between core entropy and structural gradients in galaxy clusters?* This is an ICM physics question with implications for AGN feedback modelling, thermal conduction prescriptions, and the interpretation of CC/NCC demographics in current and upcoming X-ray surveys (eROSITA, XRISM, Athena).

6. Conclusions

We present a pre-registered analysis of entropy–structure coupling in TNG-Cluster, finding:

(1) A strong negative correlation $\rho(n_e\text{-slope}, K_0) = -0.872$ ($p \sim 10^{-111}$), contrasting with the ACCEPT positive correlation $\rho(\alpha, y_{\text{CCT}}) \sim +0.36$.

(2) The sign flip is robust against mass confounding ($\rho_{\text{partial|M}} = -0.880$), consistent across all four mass bins, and preserved under proxy-based definition matching.

(3) A decisive Path A test (full Cavagnolo-model alpha-fit) is pre-registered and ready for execution, with locked parameters and stop/go criteria published prior to analysis.

These results indicate a potential sim–obs mismatch in cool-core entropy–structure coupling that warrants further investigation across multiple simulation codes and with forward-modelled mock observations.

Data Availability

All analysis scripts, pre-registered test specifications, and derived data products are available at [doi:10.6084/m9.figshare.31286368](https://doi.org/10.6084/m9.figshare.31286368). The TNG-Cluster simulation data are publicly available at www.tng-project.org/data/cluster/.

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